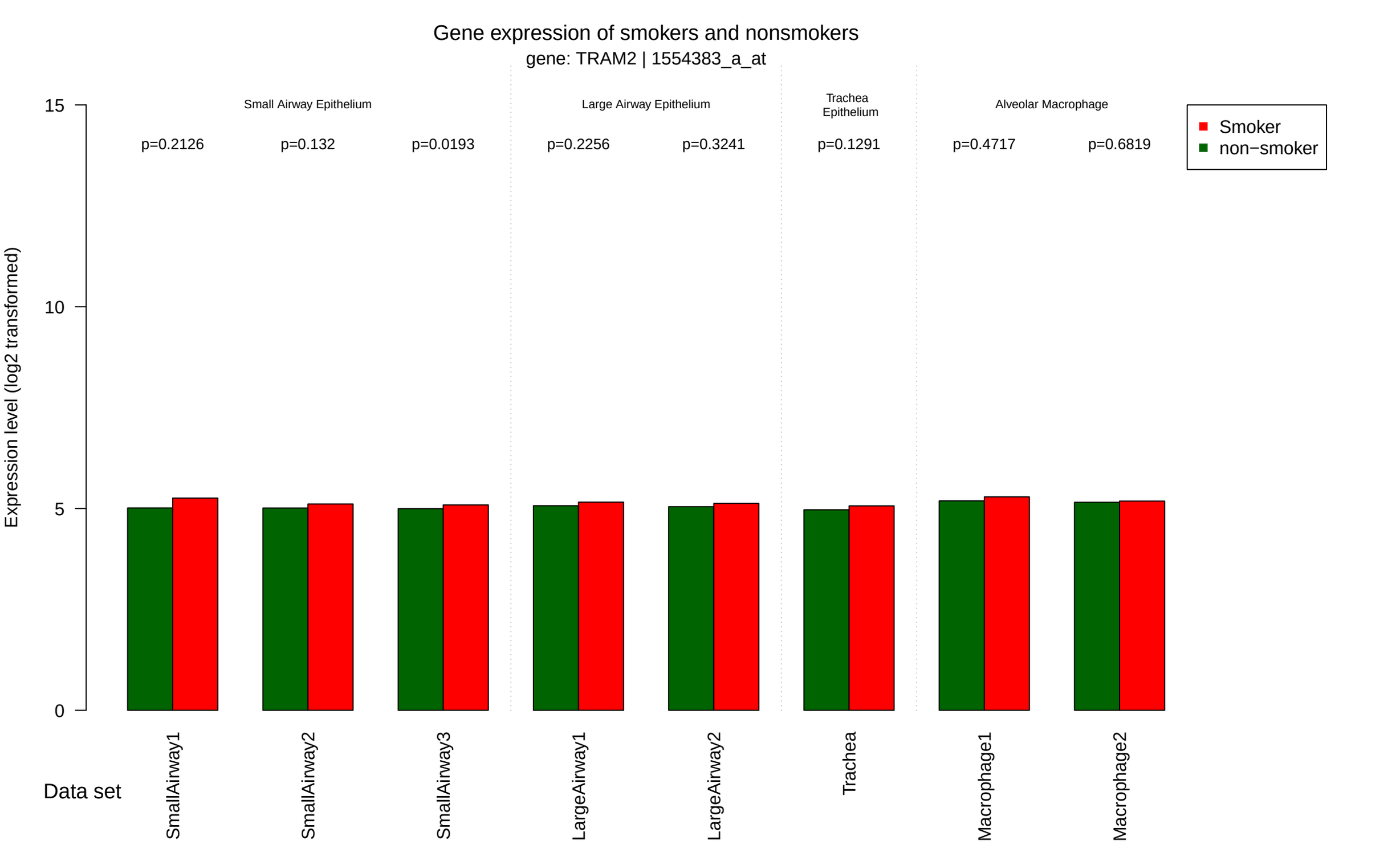
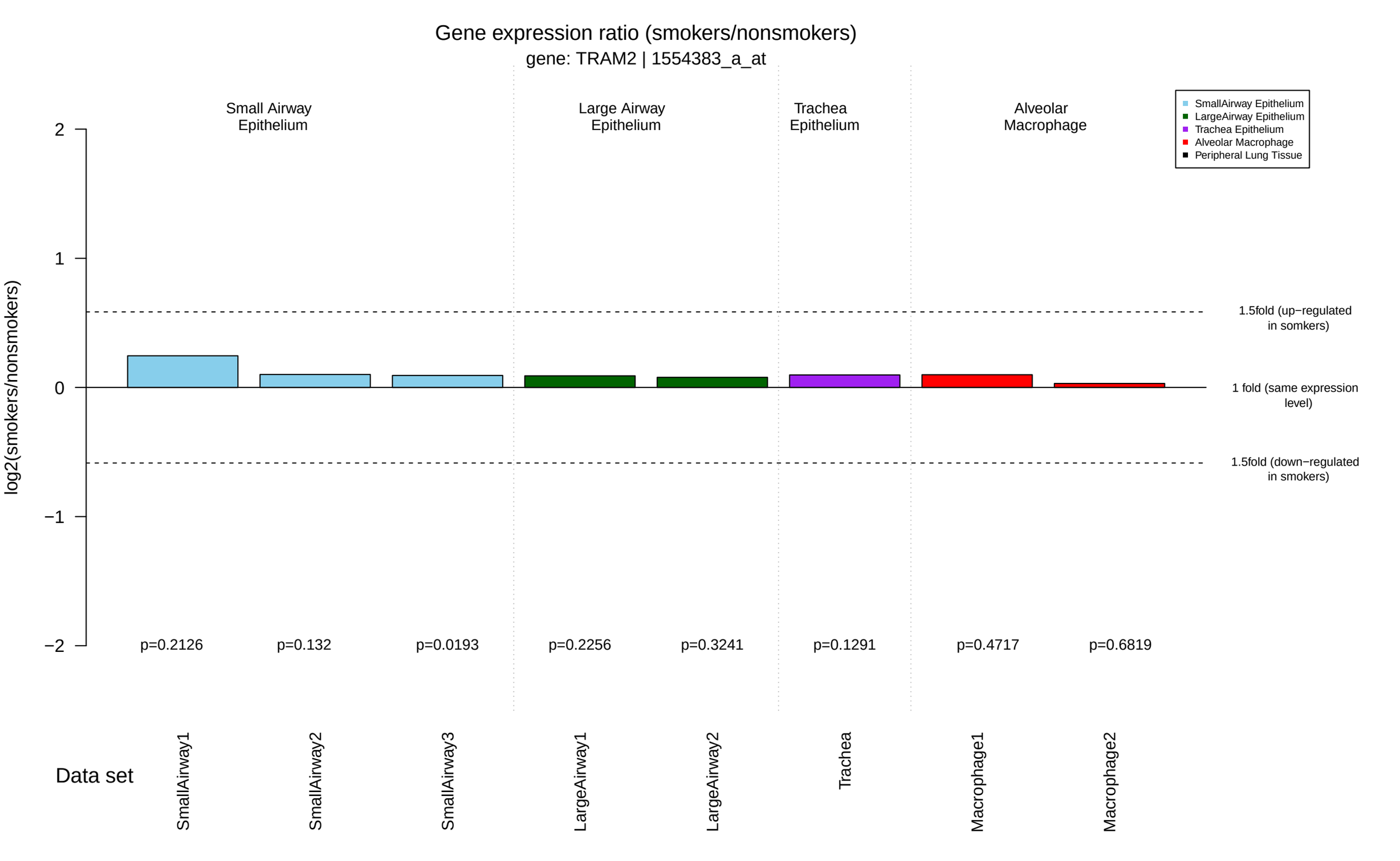
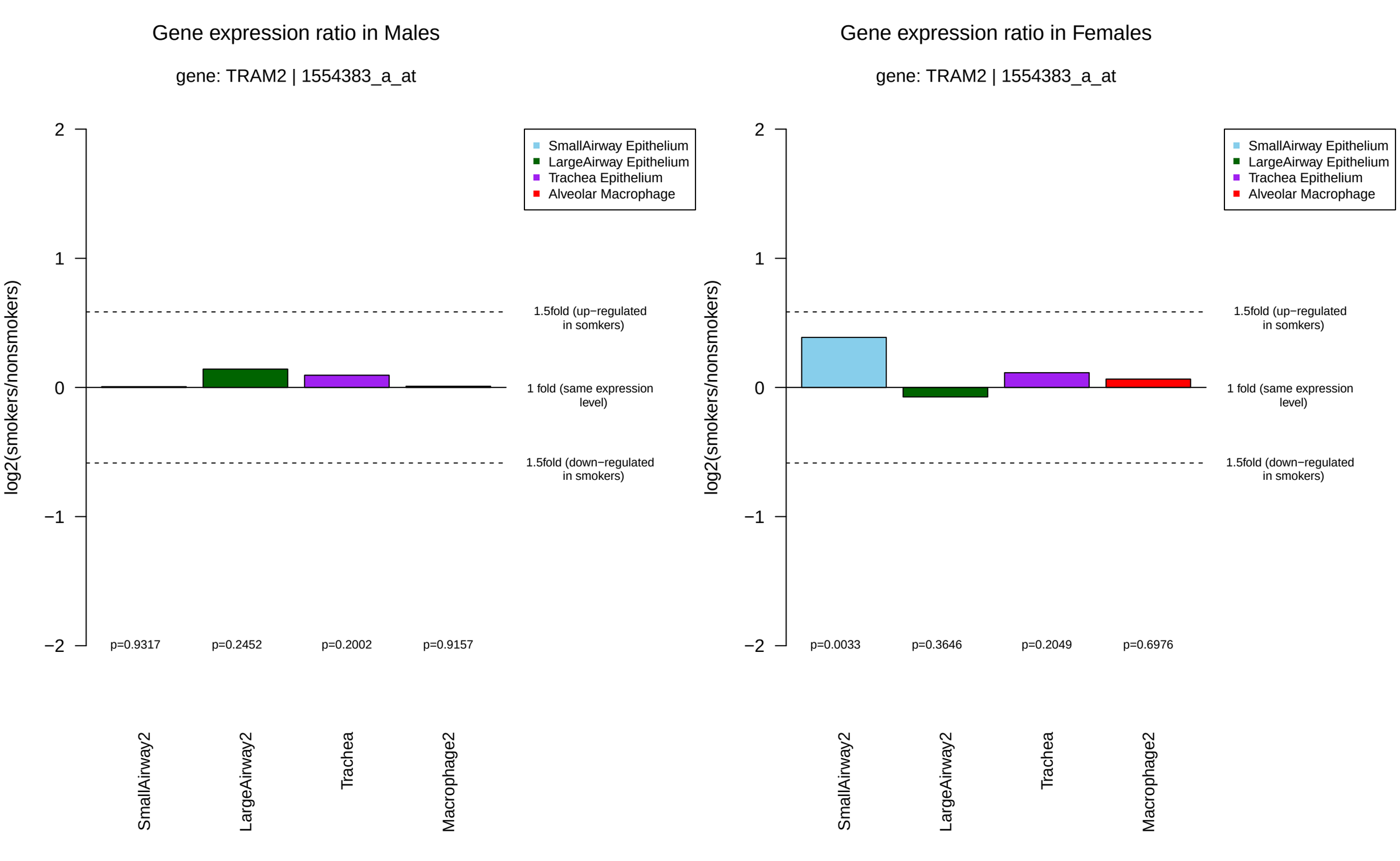
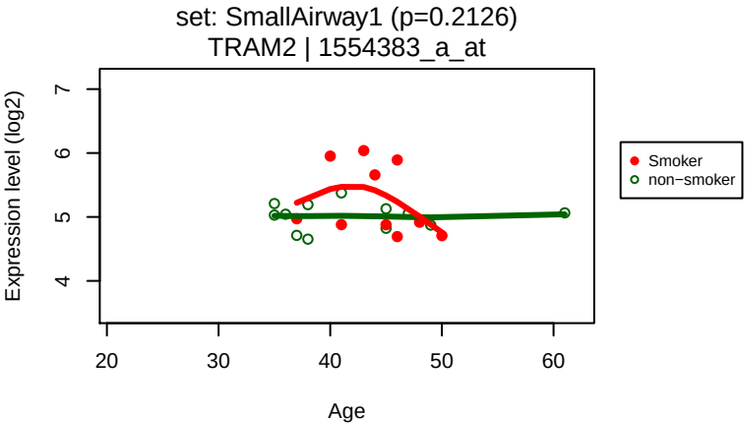
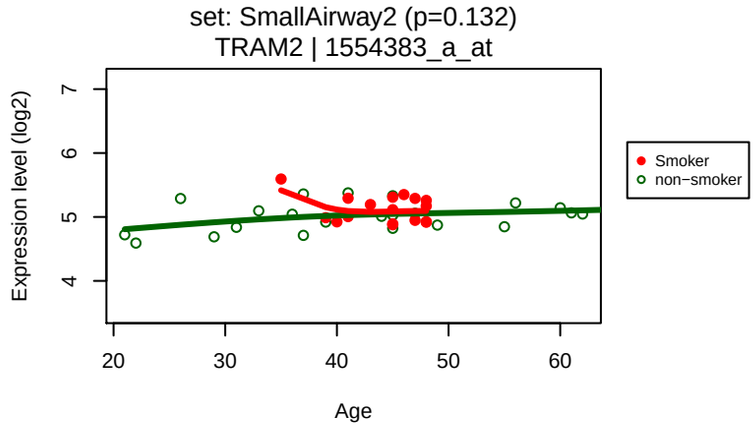
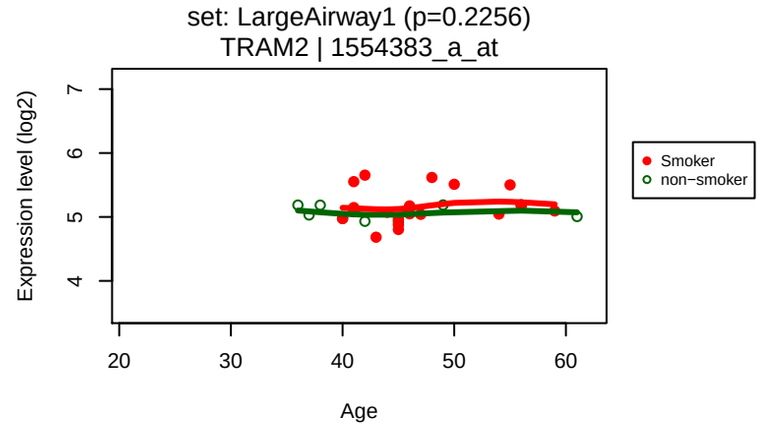

Gene expression of smokers and nonsmokers
gene: TRAM2 | 1554383_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2126
p=0.132
p=0.0193
p=0.2256
p=0.3241
p=0.1291
p=0.4717
p=0.6819
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

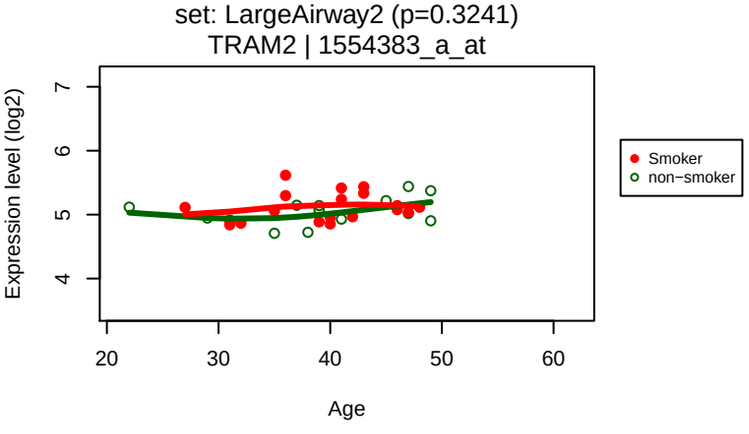

Gene expression ratio (smokers/nonsmokers)
gene: TRAM2 | 1554383_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2126
p=0.132
p=0.0193
p=0.2256
p=0.3241
p=0.1291
p=0.4717
p=0.6819
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

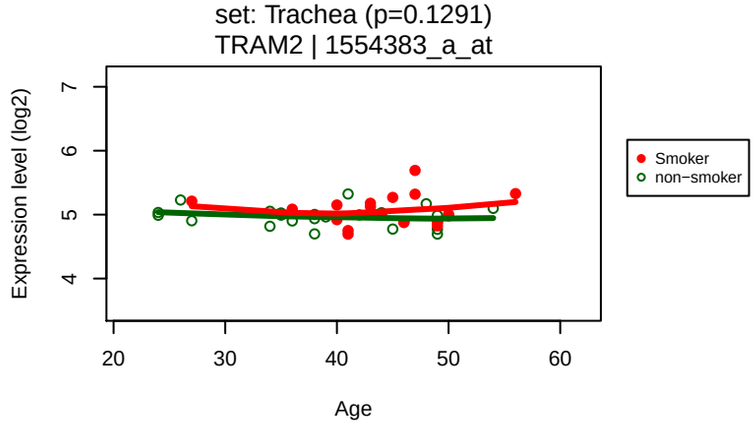

Gene expression ratio in Males
gene: TRAM2 | 1554383_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9317
p=0.2452
p=0.2002
p=0.9157
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAM2 | 1554383_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0033
p=0.3646
p=0.2049
p=0.6976
SmallAirway2
LargeAirway2
Trachea
Macrophage2

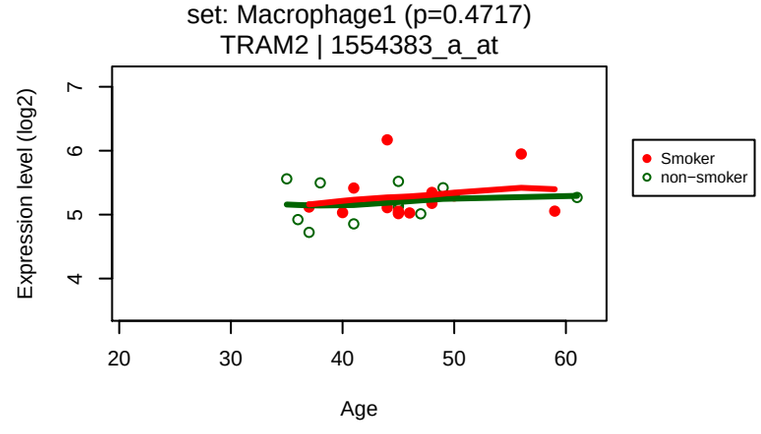

set: SmallAirway1 (p=0.2126)
TRAM2 | 1554383_a_at
Expression level (log2)
Age
Smoker
non−smoker

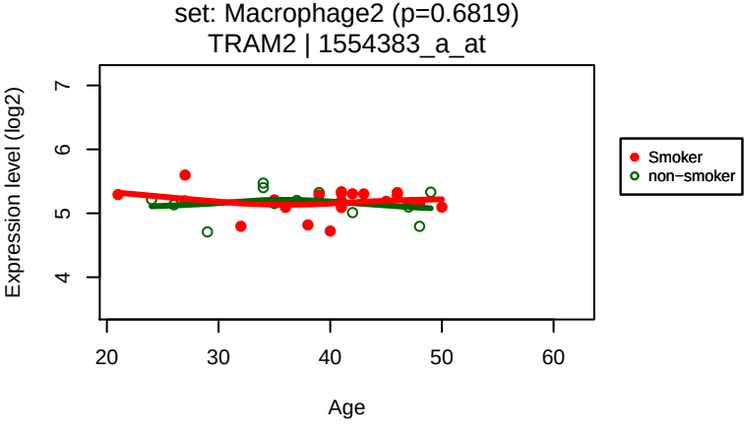

set: SmallAirway2 (p=0.132)
TRAM2 | 1554383_a_at
Expression level (log2)
Age
Smoker
non−smoker

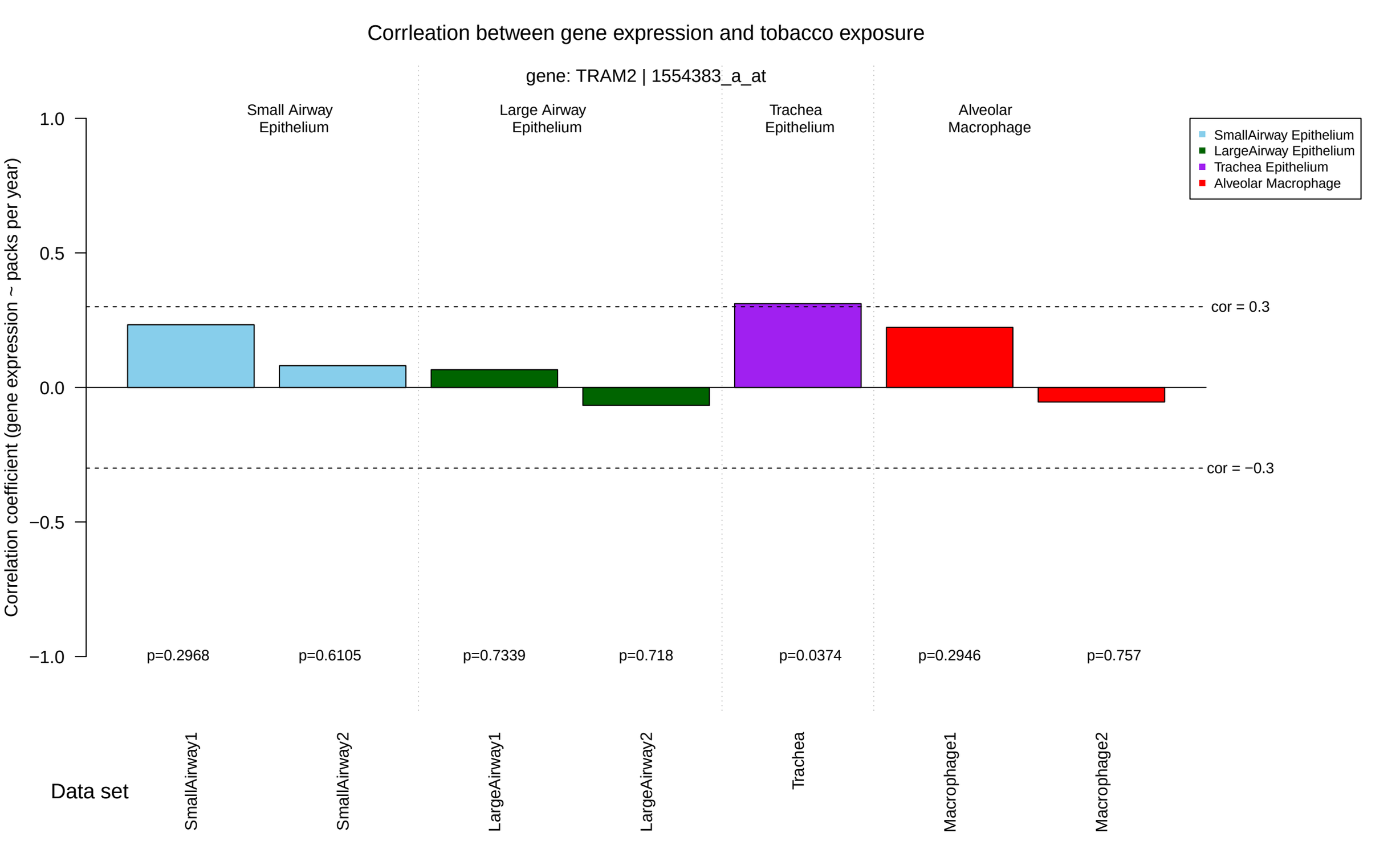

set: LargeAirway1 (p=0.2256)
TRAM2 | 1554383_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3241)
TRAM2 | 1554383_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1291)
TRAM2 | 1554383_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4717)
TRAM2 | 1554383_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6819)
TRAM2 | 1554383_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAM2 | 1554383_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2968
p=0.6105
p=0.7339
p=0.718
p=0.0374
p=0.2946
p=0.757
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2